Sadamori Kojaku

Assistant Professor at School of Systems Science and Industrial Engineering
Thomas J. Watson College of Engineering and Applied Science, Binghamton University
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Education

Ph.D. Computer Science Sep. 2013 – Sep. 2015
Hokkaido University, Japan (Thesis supervisor: Prof. Mineichi Kudo)

M.S. System Engineering Apr. 2010 – Mar. 2012
Hokkaido University, Japan (Thesis supervisor: Prof. Hajime Igarashi)

B.S. System Engineering Apr. 2007 – Mar. 2010
Hokkaido University, Japan (Thesis supervisor: Prof. Hajime Igarashi)

Experience

Assistant Professor Aug. 2023 – Present
Systems Science and Industrial Engineering, Binghamton University *Binghamton, USA*

- Conducting research in the field of complex systems and network science
- Supervising Three Ph.D. students in their dissertations and research projects
- Teaching two graduate courses in Systems Science and Industrial Engineering

Postdoctoral Research Fellow Feb. 2020 – Jul. 2023
School of Informatics, Computing, and Engineering, Indiana University *Bloomington, USA*

- Developed algorithms for analyzing large-scale data on scholarly data
- Mentored one Master and several Ph.D. students in their research projects
- Teaching a graduate course on Data Visualization

Specially Appointed Professor (Postdoc) Apr. 2019 – Jan. 2020
Research Institute for Economics and Business Administration, Kobe University *Kobe, Japan*

- Developed a search engine for finding relevant reviewers for a paper (patented)
- Mentored one Ph.D student in developing their research proposals and methodologies

Research Associate Apr. 2016 – Mar. 2019
Department of Engineering Mathematics, University of Bristol *Bristol, UK*

- Conducted research on complex networks

Manuscript in preparation

(BU students underlined; my name in italics)

[A1] Yoshiaki Fujita, Akshay Gangadhar, *Sadamori Kojaku*. On the high-order cumulative advantage in citation and collaboration networks in science.

[A2] Mario Franco, *Sadamori Kojaku*, Carlos Gershenson. Is vanilla is enough, why do we need chocolate and strawberry? A Neapolitan approach to artificial neural networks.

[A3] Dheeraj Tommondru, Xuanchi Li, Xin Wang, *Sadamori Kojaku*. Feature selection method based on the adversarial dismantling of feature networks.

[A4] Xuanchi Li, *Sadamori Kojaku*. Fast inference on the maximum entropy model for networks.

Peer reviewed journal papers

[B1] Munjung Kim, *Sadamori Kojaku*, Yong-Yeol Ahn. Uncovering simultaneous breakthroughs with a robust measure of disruptiveness. *Science Advances*, 2026.

[B2] *Sadamori Kojaku**[1], Robert Mahari*, Sandro Claudio Lera, Esteban Moro, Alex Pentland, Yong-Yeol Ahn. Community-centric modeling of citation dynamics explains collective citation patterns in science, law, and patents. *Nature Communications*, 2026.

[B3] Filipi N. Silva, *Sadamori Kojaku*, Alessandro Flammini, Filippo Radicchi, Santo Fortunato. Scale invariance and statistical significance in complex weighted networks. *Physical Review E*, 113, 034310, 2026.

[B4] Filippo Radicchi, Filipi N. Silva, Alessandro Flammini, Santo Fortunato, *Sadamori Kojaku*. Detectability threshold in weighted modular networks. *Physical Review E*, 113, 014318, 2026.

[B5] Attila Varga, *Sadamori Kojaku*, Filipi Silva. Measuring Research Interest Similarity with Transition Probabilities. *Quantitative Science Studies*, 1–40, 2025.

[B6] Isabel Constantino, *Sadamori Kojaku*, Santo Fortunato, Yong-Yeol Ahn. Representing the Disciplinary Structure of Physics: A Comparative Evaluation of Graph and Text Embedding Methods. *Quantitative Science Studies*, 6, 263–280, 2025.

[B7] Yu Tian, *Sadamori Kojaku*, Hiroki Sayama, Renaud Lambiotte. Matrix-Weighted Networks for Modeling Multidimensional Dynamics: Theoretical Foundations and Applications to Network Coherence. *Physical Review Letters*, 134, no. 23, 237401, 2025.

[B8] *Sadamori Kojaku*, Filippo Radicchi, Yong-Yeol Ahn, Santo Fortunato. Network community detection via neural embeddings. *Nature Communications*, 15, no. 1, 9446, 2024.

[B9] Bianka Kovács, *Sadamori Kojaku*, Gergely Palla, Santo Fortunato. Iterative embedding and reweighting of complex networks reveals community structure. *Scientific Reports*, 17184, 2024.

[B10] Dakota Murray, Jisung Yoon, *Sadamori Kojaku*, Rodrigo Costas, Woo-Sung Jung, Staša Milojević, Yong-Yeol Ahn. Unsupervised embedding of trajectories captures the latent structure of mobility. *PNAS*, 2023.

[B11] *Sadamori Kojaku*, Giacomo Livan, Naoki Masuda. Detecting anomalous citation groups in journal networks. *Scientific Reports*, 11, 14524, 2021.

[B12] *Sadamori Kojaku*, Laurent Hébert-Dufresne, Enys Mones, Sune Lehmann, Yong-Yeol Ahn. The effectiveness of backward contact tracing in networks. *Nature Physics*, 1745-2481, 2021.

[B13] *Sadamori Kojaku*, Naoki Masuda. Constructing networks by filtering correlation matrices: A null model approach. *Proceedings of the Royal Society A*, 475, 2231, 2019.

[B14] *Sadamori Kojaku*, Mengqiao Xu, Haoxiang Xia, Naoki Masuda. Multiscale core-periphery structure in a global liner shipping network. *Scientific Reports*, 9, 404, 2019.

[B15] *Sadamori Kojaku*, Giulio Cimini, Guido Caldarelli, Naoki Masuda. Structural changes in the interbank market across the financial crisis from multiple core-periphery analysis. *Journal of Network Theory in Finance*, 4, 33-51, 2018.

[1] Equal contribution

[B16] Naoki Masuda, *Sadamori Kojaku*, Yukie Sano. A configuration model for correlation matrices. *Physical Review E*, 98, 012312, 2018.

[B17] *Sadamori Kojaku*, Naoki Masuda. A generalised significance test for individual communities in networks. *Scientific Reports*, 8, 7351, 2018.

[B18] *Sadamori Kojaku*, Naoki Masuda. Core-periphery structure requires something else in the network. *New Journal of Physics*, 20, 043012, 2018.

[B19] *Sadamori Kojaku*, Naoki Masuda. Finding multiple core-periphery pairs in networks. *Physical Review E*, 96, 052313, 2017.

[B20] *Sadamori Kojaku*, Ichigaku Takigawa, Mineichi Kudo, Hideyuki Imai. Dense core model for cohesive subgraph discovery. *Social Networks*, 44, 143-152, 2016.

[B21] *Sadamori Kojaku*, Kota Watanabe, Hajime Igarashi. Rational Forgettable Profit Sharing Reinforcement Learning. *IEEJ*, 3, 448-454, 2012. Original in Japanese.

[B22] *Sadamori Koujaku*, Kota Watanabe, Hajime Igarashi. Adaptive profit sharing reinforcement learning for dynamic environment. , 2011. Listed under journal papers in original CV.

Peer reviewed full conference papers

[C1] Xuanchi Li, Xin Wang, *Sadamori Kojaku*. Fast Unbiased Sampling of Networks with Given Expected Degrees and Strengths. Proceedings | Complex Networks and their Applications 2025, Vestal, New York, 2025.

[C2] Yeunkyung Cho, Sarah Escotto-Rodríguez, *Sadamori Kojaku*. Emotion contagion through emotion management. Proceedings | Complex Networks and their Applications 2025, Vestal, New York, 2025.

[C3] Xin Wang, Stephanie Tulk Jesso, *Sadamori Kojaku*, David M Neyens, Min Sun Kim. VizTrust: A Visual Analytics Tool for Capturing User Trust Dynamics in Human-AI Communication. Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems , pages 1–10, 2025.

[C4] Rachith Aiyappa, Xin Wang, Munjung Kim, Ozgur Can Seckin, Yong-Yeol Ahn, *Sadamori Kojaku*. Implicit degree bias in the link prediction task. Forty-second International Conference on Machine Learning, 2025.

[C5] Artin Tonekaboni, Xin Wang, *Sadamori Kojaku*, Luis M Rocha. BingAster at #SMM4H-HeaRD 2025: Identifying Dementia Caregivers on Twitter Using Prompt-Based LLMs and Cognitive Distortion Patterns. , 2025.

[C6] *Sadamori Kojaku*, Jisung Yoon, Isabel Constantino, Yong-Yeol Ahn. Residual2Vec: Debiasing graph embedding with random graphs. NeurIPS, 2021 Acceptance rate 26%.

[C7] Xia Cui, *Sadamori Kojaku*, Naoki Masuda, Danushka Bollegala. Solving feature spareness in text classification using core-periphery decomposition. Proceedings of the 7th Joint Conference on Lexical and Computational Semantics , New Orleans, USA, pages 225-264, 2018.

[C8] *Sadamori Koujaku*, Mineichi Kudo, Ichigaku Takigawa, Hideyuki Imai. Community change detection in dynamic networks in noisy environment. The 24th International Conference on World Wide Web, Florence, Italy, 2015.

[C9] *Sadamori Koujaku*, Mineichi Kudo, Ichigaku Takigawa, Hideyuki Imai. Structural change point detection for social networks. The World Congress on Engineering, London, the United Kingdom, 2013.

Commentary

[D1] *Sadamori Kojaku*. New developments in network science through embedding methods. Special Issue "Frontiers of Complex Network Research". Journal of the Society of Instrument and Control Engineers, 65, 5, 185-191 (2021)

Invited Talks

[E1] *Sadamori Kojaku*. Machine learns simplicity in complexity, Complex Systems Research Exchange, July 2, 2024.

[E2] *Sadamori Kojaku*. Keynote Speaker. 17th International Conference on Complex Networks (CompleNet 2026), CompleNet 2026, Zaragoza, Spain, May 4-8, 2026.

[E3] Invited Member. Project: Modeling Higher-Order Structures in Complex Networks and Elucidating Influence Propagation and Robustness Between Remote Nodes (PI: Yasuko Yamano), 2026.

[E4] *Sadamori Kojaku*. Neural embeddings unveil simplicity in complex systems, American Physical Society March Meeting, Minneapolis, MN, USA, March 4-8, 2024.

[E5] *Sadamori Kojaku*. Unsupervised embedding of trajectories captures the latent structure of scientific migration, CoCo Seminar, Center for Collective Dynamics of Complex Systems, Binghamton University , USA, January 24, 2024.

[E6] *Sadamori Kojaku*. Distilling rich but crude scholarly data using representation learning, IUNI Lunch Colloquium: Science of Science and Networks, Indiana, USA, October 28, 2022.

[E7] *Sadamori Kojaku*, Jisung Yoon, Isabel Constantino, Yong-Yeol Ahn. Residual2Vec: Debiasing graph embedding with random graphs, Network Inequality, International School and Conference on Network Science (NetSci) 2022 , Shanghai, China, July 21, 2022.

[E8] *Sadamori Kojaku*. Algorithms for detecting network cores and their applications, Network Science Seminar, Institute of Statistical Mathematics, Japan, August 28-30, 2019.

[E9] *Sadamori Kojaku*, Laurent Hébert-Dufresne, Enys Mones, Sune Lehmann, Yong-Yeol Ahn. The effectiveness of backward contact tracing in networks, The State University of New York at Buffalo, June 4, 2021.

Oral Presentations & Poster Presentations (Refereed)

(BU students underlined; my name in italics)

Oral Presentations

[F1] Filippo Radicchi, Filipi N. Silva, Alessandro Flammini, Santo Fortunato, *Sadamori Kojaku*. Detectability threshold in weighted modular networks, International School and Conference on Network Science (NetSci) , Boston, MA, USA, June 1-5, 2026.

[F2] Yoshiaki Fujita, Akshay Gangadhar, *Sadamori Kojaku*. On the high-order cumulative advantage in citation and collaboration networks in science, NetSci, Maastricht; NERCCS, April 2025.

[F3] *Sadamori Kojaku*, Filippo Radicchi, Yong-Yeol Ahn, Santo Fortunato. Network community detection via neural embeddings, CompleNet, Casa José de Alencar, Brazil, April 2025.

[F4] Rachith Aiyappa, Xin Wang, Munjung Kim, Ozgur Can Seckin, Jisung Yoon, Yong-Yeol Ahn, *Sadamori Kojaku*. Implicit degree bias in the link prediction task, CompleNet, Casa José de Alencar, Brazil, April 2025.

[F5] Rachith Aiyappa, Xin Wang, Munjung Kim, Ozgur Can Seckin, Jisung Yoon, Yong-Yeol Ahn, *Sadamori Kojaku*. Implicit degree bias in the link prediction task, NetSci, Maastricht, the Netherlands, June 2025.

[F6] Xuanchi Li, Xin Wang, *Sadamori Kojaku*. Near-linear time algorithm for the configuration models for networks, CompleNet, Casa José de Alencar, Brazil, April 2025.

[F7] Xuanchi Li, Xin Wang, *Sadamori Kojaku*. Near-linear time algorithm for the configuration models for networks, NetSci, Maastricht, the Netherlands, June 2025.

[F8] *Sadamori Kojaku*. Unsupervised Embedding of Trajectories Captures the Latent Structure of Scientific Migration, CCSS Workshop on Computational Social Science: Methods and Applications, Center for Computational Social Science, Kobe University , Japan, December 26, 2024.

[F9] *Sadamori Kojaku*, Robert Mahari, Sandro Lera, Esteban Moro, Alex Pentland, Yong-Yeol Ahn. Uncovering the universal dynamics of citation systems: From science of science to law of law and patterns of patents, NERCCS, Clarkson, NY, USA, March 20-22, 2024.

[F10] *Sadamori Kojaku*, Robert Mahari, Sandro Lera, Esteban Moro, Alex Pentland, Yong-Yeol Ahn. Uncovering the universal dynamics of citation systems: From science of science to law of law and patterns of patents, International School and Conference on Network Science (NetSci) , Vienna, Austria, Jul 12-14, 2023.

[F11] *Sadamori Kojaku*, Robert Mahari, Sandro Lera, Esteban Moro, Alex Pentland, Yong-Yeol Ahn. Uncovering the universal dynamics of citation systems: From science of science to law of law and patterns of patents, ICSSI, Chicago, IL, USA, June 26-29, 2023.

[F12] *Sadamori Kojaku*, Filippo Radicchi, Yong-Yeol Ahn, Santo Fortunato. Network community detection via neural embeddings, International School and Conference on Network Science (NetSci) , Vienna, Austria, Jul 12-14, 2023.

[F13] *Sadamori Kojaku*, Clara Boothby, Filipi Nascimento Silva, Attila Varga, Xiaoran Yan, Staša Milojević, Alessandro Flammini, Filippo Menczer, Yong-Yeol Ahn. Maping Scientific Foraging, ICSSI, Washington D.C., USA, 6-9 June 2022.

[F14] *Sadamori Kojaku*, Xiaoran Yan, Jisung Yoon, Filipi N. Silva, Vincent Lariviere, Yong-Yeol Ahn. DisamBERT: author name disambiguation with BERT, ICSSI, Washington D.C., USA, 6-9 June 2022.

[F15] *Sadamori Kojaku*, Laurent Hébert-Dufresne, Enys Mones, Sune Lehmann, Yong-Yeol Ahn. The effectiveness of backward contact tracing in networks, International School and Conference on Network Science (NetSci) , Virtual, 05-10 July 2021.

[F16] *Sadamori Kojaku*, Jisung Yoon, Yong-Yeol Ahn. Residual2Vec: A null model approach for graph embedding, International School and Conference on Network Science (NetSci) , Virtual, 05-10 July 2021.

[F17] Dakota Murray, Jisung Yoon, *Sadamori Kojaku*, Rodrigo Costas, Woo-Sung Jung, Staša Milojević, Yong-Yeol Ahn. Unsupervised embedding of trajectories captures the latent structure of mobility, International School and Conference on Network Science (NetSci) , Virtual, 05-10 July 2021.

[F18] *Sadamori Kojaku*, Attila Varga, Xiaoran Yan, Filipi N. Silva, Staša Milojević, Alessandro Flammini, Yong-Yeol Ahn. The landscape of the COVID-19 research: A neural embedding approach, International School and Conference on Network Science (NetSci) , Rome, Italy, 17-25 September 2020.

[F19] *Sadamori Kojaku*, Giacomo Livan, Naoki Masuda. Detecting citation cartels in journal networks, International School and Conference on Network Science (NetSci) , Rome, Italy, 17-25 September 2020.

[F20] *Sadamori Kojaku*, Giulio Cimini, Guido Caldarelli, Naoki Masuda. Structural changes in the interbank market across the financial crisis from multiple core-periphery analysis, International School and Conference on Network Science (NetSci) , Vermont, U.S., May 26-31 2019.

[F21] *Sadamori Kojaku*, Naoki Masuda. Core-periphery structure in degree-heterogeneous networks, International School and Conference on Network Science (NetSci-X) , Hangzhou, China, 2018.

[F22] *Sadamori Kojaku*, Naoki Masuda. Finding multiple core-periphery structure with random walks, 5th International Workshop on Complex Networks and their Applications , Milan, Italy, November 30 – December 2 2016.

[F23] Keigo Kimura, Mineichi Kudo, Lu Sun, *Sadamori Kojaku*. Fast random k-labelsets for large-scale multi-label classification, 23rd International Conference on Pattern Recognition, Cancun, Mexico, December 4-8 2016.

Poster Presentations

[F24] Xuanchi Li, *Sadamori Kojaku*. Fast inference for maximum entropy configuration models, International School and Conference on Network Science (NetSci) , Boston, MA, USA, June 1-5, 2026.

[F25] Rachith Aiyappa, Xin Wang, Munjung Kim, Ozgur Can Seckin, Jisung Yoon, Yong-Yeol Ahn, *Sadamori Kojaku*. Implicit degree bias in the link prediction task, Forty-second International Conference on Machine Learning (ICML) , Vienna, Austria, July 21-27, 2025.

[F26] Yoshiaki Fujita, Akshay Gangadhar, *Sadamori Kojaku*. On the high-order cumulative advantage in citation and collaboration networks in science, NetSci, Maastricht, the Netherlands, June 2025.

[F27] Xin Wang, *Sadamori Kojaku*. User Trust Modeling in Conversational User Interface Based on Word Embedding Bias, The ACM Conference on Conversational User Interfaces, July 8-10, 2024.

[F28] Govind Gandhi, Yong-Yeol Ahn, *Sadamori Kojaku*. Self-Supervised Modularity Maximization using graph embeddings for clustering, International School and Conference on Network Science (NetSci) , Quebec, Canada, June 16-21, 2024.

[F29] Xin Wang, *Sadamori Kojaku*. Analyzing Patient Reviews on Google Map Hospital Profiles through Neural Embedding and Network Modeling, NERCSS, Clarkson, NY, USA, March 20-22, 2024.

[F30] Ashutosh Tiwari, *Sadamori Kojaku*, Yong-Yeol Ahn. A biased contrastive learning debiases graph neural networks, International School and Conference on Network Science (NetSci) , Vienna, Austria, Jul 12-14, 2023.

[F31] *Sadamori Kojaku*, Clara Boothby, Filipi Nascimento Silva, Attila Varga, Xiaoran Yan, Staša Milojević, Alessandro Flammini, Filippo Menczer, Yong-Yeol Ahn. Understanding the landscape of COVID-19 research by using neural embedding, ICSSI, Chicago, IL, USA, June 26-29, 2023.

[F32] Munjung Kim, *Sadamori Kojaku*, Yong-Yeol Ahn. Quantifying disruptiveness using a neural embedding method, ICSSI, Chicago, IL, USA, June 26-29, 2023.

[F33] *Sadamori Kojaku*, Naoki Masuda. Constructing networks from correlation matrices: An application to economical data, Threshold Networks, Nottingham, UK, July 22-24, 2019.

[F34] *Sadamori Kojaku*, Naoki Masuda. A generalised significance test for individual communities in networks, International School and Conference on Network Science (NetSci) , Paris, France, June 11-15, 2018.

[F35] *Sadamori Kojaku*, Naoki Masuda. Multi-scale organisation of core-periphery structure in networks, 1st Latin American Conference on Complex Networks, Puebla, Mexico, September 25-29, 2017.

[F36] *Sadamori Kojaku*, Naoki Masuda. Core-periphery structure of networks: Consideration for random heterogeneous networks, International School and Conference on Network Science (NetSci) , Indianapolis, Indiana, USA, 2017.

[F37] *Sadamori Kojaku*, Naoki Masuda. An extension of modularity for finding multiple core/periphery structure in networks, International School and Conference on Network Science (NetSci-X) , Tel Aviv, Israel, January 15-18, 2017.

Grants

[G1] PI: *Sadamori Kojaku*. Co-PI: Bryan Acton. Project: TEAMS: Team Engineering for Industrial AI Systems. Funded by SUNY-IBM AI Research Alliance, Feb. 2026 – Jul. 2027. $100,000 USD.

[G2] PI: *Sadamori Kojaku*. Co-PI: Guido Caldarelli, Daigo Uemoto, Takashi Kamihigashi. Project: Correlation-based reconstruction of financial networks for systemic risk control. Funded by JSPS Bilateral Exchange Program, 2020. 8,000,000 JPY Withdrawn due to transition to a U.S. institution.

Patents

[H1] *Sadamori Kojaku*, Takashi Kamihigashi. Academic paper reviewer search device, reviewer search method, and reviewer search program. 2024 [7470369. Japan Patent Office]

[H2] *Sadamori Kojaku*, Takayuki Osogami. Prediction method, prediction system and program. 2013 [9087294. Japan Patent Office]

Honors

- Best Presentation Award. The International Conference on Complex Networks (CompleNet). 2025
- Art of Science Competition Finalist, Judges Choice [14/85 entries]. Binghamton University. 2024
- Outstanding Faculty Award from Students with Disabilities. Binghamton University. 2024
- Best Contribution on Financial Networks Award [1/58 presenters]. International School and Conference on Network Science (NetSci-X). 2017
- Dean Award. Graduate School of Information Science and Technology, Hokkaido Univ.. 2015
- Best Student Award. The World Congress on Engineering. 2013

Teaching

SSIE 419/519	Applied Soft Computing. Instructor (Spring 2024). Binghamton University
SSIE 419/519	Applied Soft Computing. Instructor (Spring 2024). Binghamton University
SSIE 641	Advanced Topics in Network Science. Instructor (Fall 2024). Binghamton University
SSIE 641	Advanced Topics in Network Science. Instructor (Fall 2023). Binghamton University
BL-INFO-I590	Data Visualization. Instructor (Fall 2023). Indiana University

PhD Student Guidance Committee

(As Advisor or Co-Advisor)

Chand Sahil Mansuri (SS), Expected to Graduate in 2027.

Xin (Vision) Wang (SS), Expected to Graduate in 2027.

Luke Netto (SS), Expected to Graduate in 2028.

Hayford Adjavor (SS), Expected to Graduate in 2026.

Sarah Alzubi

Amogh Kokari

MS Student Guidance Committee

(As Advisor or Committee Member)

Xuanchi Li (SS). Advisor. Expected to Graduate in 2027.

Miriam Flores (SS). The Impact of Autonomous Vehicles on Traffic Flow. Expected to Graduate in 2024.

International and Professional Services

[H3] Chair of Rep4CS, a satellite workshop on Representation learning for complex systems, as a part of Conference on Complex Systems 2024 at Exeter in UK.

[H4] Organizer of NetSci-X 2020, International School and Conference on Network Science in Tokyo. 2020.

[H5] Program Committee Member in International School and Conference on Network Science (NetSci) (2019, 2020, 2021, 2022, 2023).

[H6] Referee for Nature Human Behavior; Nature Communications; Scientific Reports; Journal of Complex Networks; Journal of Computational Social Science; PLOS ONE; ECAI

Service within the University

[I1] Faculty Fellow, Bernard M. and Ruth R. Bass Center for Leadership Studies, Binghamton University. 2026–present.

[I2] Talk at DataViz Community Presentations and Networking. Gravity of Ideas: Mapping Science at BU. Spring 2025.

[I3] Talk at DataViz Community Presentations and Networking. Unsupervised embedding of trajectories captures the latent structure of scientific migration. Fall 2024.

[I4] Talk at Data Salon. Machine learns simplicity from complexity. Nov 17, 2023.

[I5] Talk at DataViz Pi Day 2024 Presentations and Networking. A Landscape of All Sciences: Neural Networks of Over 400M Publications. March 14, 2023.